

Laplacian closability: the finite-exponent theorem and the strong- L^∞ counterexample

Full resolution of the expectation in arXiv:2311.08058

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Status. Candidate counterexample and full resolution, likely valid, awaiting human review. The expected necessary condition holds for every finite output exponent in all dimensions, but fails at the strong L^∞ endpoint.

1 Source expectation

Alberti, Bate, and Marchese study closability of differential operators with respect to Radon measures in [1]. In concluding Remark 1.7(iv), page 4, they suggest that closability of the Laplacian from $L^p(\mu)$ to $L^q(\mu)$ should force $\mu \ll \mathcal{L}^d$.

does not imply the closability of the gradient operator from $L^q(\mu)$ to $L^p(\mu)$, not even in dimension $d = 1$. For example, let μ be the restriction of the Lebesgue measure $\mathcal{L}^1$ to a totally disconnected compact subset of $\mathbb{R}$; then it is easy to prove that the derivative is not closable from $L^q(\mu)$ to $L^p(\mu)$ for any $1 \leq p, q \leq \infty$.

A precise characterization of the (absolutely continuous) measures μ on $\mathbb{R}$ such that the derivative is closable from $L^2(\mu)$ to $L^2(\mu)$ has been given in [4, Theorem 2.2] (see also [15, Theorem 3.1.6]).

(iv) Theorem 1.3 can be easily extended to more general *first order* differential operators than just directional derivatives, thus obtaining a statement that includes Corollaries 1.5 and 1.6 as particular cases (see Remark 3.2).

However, it seems that *second order differential operators* are not easily included in our analysis. For instance, one would guess that the Laplace operator (defined on $C_c^2(\mathbb{R}^d)$) should be closable from $L^p(\mu)$ to $L^q(\mu)$ only if μ is absolutely continuous with respect to $\mathcal{L}^d$, but such statement does not seem to follow from any of the results in this paper (at least, not easily).

(v) At the core of the proof of Theorem 1.3(ii), and of the “only if” part of Corollaries 1.5 and 1.6, are the so-called “width functions”, which are taken from

Here closability is sequential: for a linear operator it is enough to require that

$$u_n \longrightarrow 0 \text{ in the input space,} \quad \Delta u_n \longrightarrow w \text{ in the output space} \quad \implies \quad w = 0.$$

2 Idea of the proof

For finite output exponent, closability forces the Banach adjoint to have a dense domain. Every adjoint test function φ then gives a measure $\nu = \varphi\mu$ whose distributional Laplacian is also a Radon measure. The fundamental solution of the Laplacian shows that such a ν must have a Lebesgue density. Closedness plus density of the adjoint domain then forces every compact restriction of μ to have a density.

The L^∞ endpoint behaves differently. For $\mu = \mathcal{L}^d + \delta_0$, the $L^\infty(\mu)$ norm of a continuous function is its ordinary supremum norm. Thus convergence of the Laplacians is uniform. Input convergence makes the original functions converge to zero as distributions, so the uniform limit of their Laplacians must be zero. The operator is closable although μ has an atom.

3 Full dichotomy

Theorem 1. *Let $d \geq 1$, let μ be a positive Radon measure on $\mathbb{R}^d$, and initially define Δ on $C_c^2(\mathbb{R}^d)$.*

1. *If $1 \leq p \leq \infty$ and $1 \leq q < \infty$, closability of*

$$\Delta : L^p(\mu) \supset C_c^2(\mathbb{R}^d) \longrightarrow L^q(\mu)$$

forces $\mu \ll \mathcal{L}^d$.

2. *The assertion is false for $q = \infty$. For*

$$\mu = \mathcal{L}^d + \delta_0$$

the measure μ is not absolutely continuous, but Δ is closable from $L^p(\mu)$ to $L^\infty(\mu)$ for every $1 \leq p \leq \infty$.

We use two elementary lemmas. The first is the standard adjoint criterion, included for clarity.

Lemma 1 (Adjoint density). *Let X, Y be Banach spaces and let $T : \text{Dom}(T) \subset X \rightarrow Y$ be densely defined. If T is closable, then $\text{Dom}(T^*)$ is weak-star dense in Y^* . If Y is reflexive, this linear subspace is norm dense.*

Proof. Let $G \subset X \times Y$ be the graph. If $y \in Y$ annihilates $\text{Dom}(T^*)$, then $(0, y)$ annihilates every element of $G^\perp$: indeed, the second coordinate of an element of $G^\perp$ belongs to $\text{Dom}(T^*)$. The bipolar theorem for linear subspaces gives $(0, y) \in \overline{G}$. Since $\overline{G}$ is a graph, $y = 0$. Thus the preannihilator of $\text{Dom}(T^*)$ is zero, which is exactly weak-star density. In a reflexive space, a linear subspace with weakly dense closure is norm dense by Hahn–Banach. $\square$

Lemma 2 (Measure with measure Laplacian). *If $\nu \in \mathcal{M}_{\text{loc}}(\mathbb{R}^d)$ and $\Delta\nu \in \mathcal{M}_{\text{loc}}(\mathbb{R}^d)$ as distributions, then $\nu \ll \mathcal{L}^d$.*

Proof. Write $\eta = \Delta\nu$. Fix a ball B and choose $\chi \in C_c^\infty(\mathbb{R}^d)$ equal to one on a neighborhood of B . Let E be the fundamental solution of the Laplacian and set

$$v := E * (\chi\eta).$$

The function E is locally integrable in every dimension, so Fubini's theorem shows that $v \in L_{\text{loc}}^1$. On B , $\Delta(\nu - v) = 0$. Weyl's lemma for harmonic distributions makes $\nu - v$ smooth on B . Hence ν is represented by an L_{loc}^1 function on B . Since B was arbitrary, $\nu \ll \mathcal{L}^d$. $\square$

Proof of Theorem 1, finite q . It suffices to restrict the operator to $C_c^\infty(\mathbb{R}^d)$. If two smooth functions representing the same $L^p(\mu)$ class already have different Laplacians in $L^q(\mu)$, the graph is not a graph and closability fails immediately; thus under the closability assumption the restriction is a well-defined operator.

First suppose $p < \infty$. Put $X = L^p(\mu)$ and $Y = L^q(\mu)$. Smooth compactly supported functions are dense in X . If the graph were closable, Lemma 1 would make $\text{Dom}(\Delta^*)$ weak-star dense in $Y^* = L^{q'}(\mu)$; for $q > 1$ this is norm density.

For $\varphi \in \text{Dom}(\Delta^*)$ there is $g \in L^{p'}(\mu)$ such that

$$\int_{\mathbb{R}^d} \varphi \Delta u \, d\mu = \int_{\mathbb{R}^d} g u \, d\mu \quad (u \in C_c^\infty(\mathbb{R}^d)).$$

Thus, for the locally finite signed measure $\nu = \varphi\mu$,

$$\Delta\nu = g\mu \in \mathcal{M}_{\text{loc}}(\mathbb{R}^d).$$

Lemma 2 gives $\varphi\mu \ll \mathcal{L}^d$.

Now suppose $p = \infty$. Define the same operator on the dense subspace $C_c^\infty(\mathbb{R}^d)$ of $X = C_0(\mathbb{R}^d)$. If $u_n \rightarrow 0$ uniformly, then $u_n \rightarrow 0$ in $L^\infty(\mu)$, so the assumed closability makes this C_0 -operator closable. For an adjoint-domain element φ , the Riesz representation theorem supplies a finite signed measure η with

$$\int \varphi \Delta u \, d\mu = \int u \, d\eta.$$

Hence $\Delta(\varphi\mu) = \eta$, and Lemma 2 again yields $\varphi\mu \ll \mathcal{L}^d$.

In either case, let

$$\mathcal{A} := \{\varphi \in L^{q'}(\mu) : \varphi\mu \ll \mathcal{L}^d\}.$$

For $q > 1$, $\mathcal{A}$ is norm closed: norm convergence in $L^{q'}(\mu)$ implies total-variation convergence of the associated measures on every compact set. For $q = 1$, $\mathcal{A} \subset L^\infty(\mu)$ is weak-star closed: for every Lebesgue-null Borel set N of finite μ -measure, the condition $\int_N \varphi \, d\mu = 0$ is weak-star closed, and one intersects these conditions over all such N and their Borel subsets.

The adjoint domain lies in $\mathcal{A}$ and is dense in the relevant topology, so $\mathcal{A} = L^{q'}(\mu)$. For each compact K , the function $\mathbf{1}_K$ belongs to $L^{q'}(\mu)$; therefore $\mu \ll \mathcal{L}^d$. An exhaustion by compact balls gives $\mu \ll \mathcal{L}^d$. $\square$

Proof of Theorem 1, $q = \infty$. Let $\mu = \mathcal{L}^d + \delta_0$. This is a positive Radon measure and is not absolutely continuous. For every continuous compactly supported function f ,

$$\|f\|_{L^\infty(\mu)} = \|f\|_\infty, \tag{1}$$

because the Lebesgue essential supremum of a continuous function is its ordinary supremum.

Suppose $u_n \in C_c^2(\mathbb{R}^d)$, $u_n \rightarrow 0$ in $L^p(\mu)$, and $\Delta u_n \rightarrow w$ in $L^\infty(\mu)$. If $p < \infty$, then $u_n \rightarrow 0$ in $L^p(\mathcal{L}^d)$ and hence in distributions. If $p = \infty$, (1) applied to u_n gives uniform convergence to zero, hence again distributional convergence.

By (1), the continuous functions Δu_n are uniformly Cauchy. They converge uniformly to a continuous function f representing w . Distributionally, however,

$$\Delta u_n \longrightarrow 0$$

because $u_n \rightarrow 0$ in distributions. Uniform convergence also gives $\Delta u_n \rightarrow f$ in distributions, so $f = 0$ and therefore $w = 0$. This proves closability for every p . $\square$

4 Consequences and limitations

The theorem completely resolves the necessary-condition expectation as stated across the output exponent range. The finite-exponent result includes $q = 1$ and the input endpoint $p = \infty$. The strong- L^∞ endpoint is not merely outside the proof: the implication is false there.

No converse is asserted. Even for absolutely continuous measures, closability may depend on the density and on the exponents. The theorem also does not concern weak-star convergence in L^∞ , which is a different output topology.

5 Novelty check and review recommendation

The local run indexes were searched for arXiv:2311.08058, Laplacian closability, measure Laplacians, and the strong- L^∞ endpoint. Bounded arXiv searches through 21 July 2026 used exact and close variants of “Laplacian closable Radon measure”, “absolute continuity”, and “ L^∞ closability”. They found the source paper but no later exact answer. This is not an exhaustive bibliographic claim.

Human review should focus on Lemma 2, the weak-star closure argument at $q = 1$, and the fact that full-support Lebesgue measure turns the output L^∞ norm into the uniform norm. Subject to those checks, this packet should replace the earlier one-dimensional partial result.

References

- [1] G. Alberti, D. Bate, and A. Marchese, *On the closability of differential operators*, J. Funct. Anal. 289 (2025), 111029; arXiv:2311.08058v2.
- [2] L. C. Evans, *Partial Differential Equations*, 2nd ed., Graduate Studies in Mathematics 19, American Mathematical Society, 2010.